

Unless stated otherwise, all problems are worth equal value and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

1. Use figure 0.1 to complete the following.

- a. Create a frequency distribution from figure 0.1 (approximate the values as best as possible).

mid	Weight	Freq.	$f \cdot x$
15.45	$15.3 \leq x < 15.6$	13	200.85
15.75	$15.6 \leq x < 15.9$	35	551.25
16.05	$15.9 \leq x < 16.2$	24	385.2
16.35	$16.2 \leq x < 16.5$	18	294.3
16.65	$16.5 \leq x < 16.8$	10	166.5
	total	100	1598.1

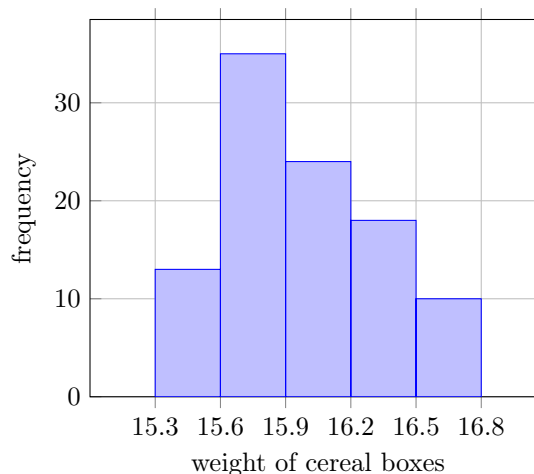


Figure 0.1: Weights of boxes of cereal.

- b. Find the mean weight of cereal boxes.

$$\bar{x} = \frac{\sum f \cdot x}{n} = \frac{1598.1}{100} = 15.981$$

- c. Find the mode weight of cereal boxes.

15.75

- d. Is the data normally distributed? Why or why not?

Not really. The mean and mode are not that close, and it is not symmetrical.

2. Forty randomly selected families were surveyed to determine the number of children in each family. The following results were found:

3.7 3.6 4.0 3.8 4.1 3.9 3.5 3.9
3.6 4.0 3.6 3.7 3.9 3.3 3.6 4.0

- a. Find the mean.

$$\bar{x} = \frac{\sum x}{n} = \frac{60.2}{16} \approx 3.7625$$

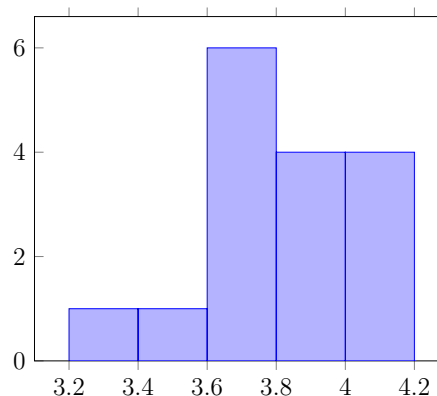
- b. Find the median.

$$\frac{3.7 + 3.8}{2} = 3.75$$

- c. Find the standard deviation.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} \approx .2217$$

- d. Construct a histogram using 5 subintervals, beginning at 3.2, and ending at 4.2.



3. Tom and Jerry golfed five times during their vacation. Their scores are given in the table below:

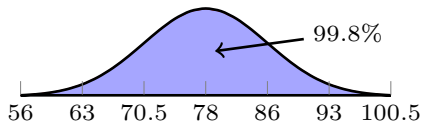
Tom	103	99	107	93	92
Jerry	101	92	83	96	111

Who is the more consistent golfer?

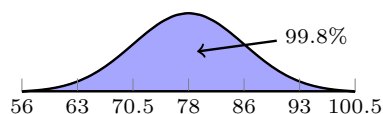
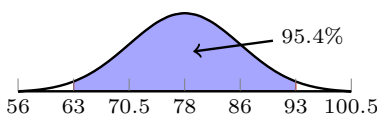
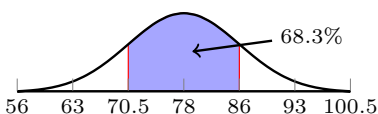
Tom has a standard deviation of $s_T \approx 6.4187$

Jerry has a standard deviation of $s_J \approx 10.4067$. So Tom is more consistent since $6.41 < 10.40$.

4. A population is normally distributed with mean 78 and standard deviation 7.5.
- a. Draw a sketch of the bell curve labeling the mean and the intervals representing 1, 2, and 3 standard deviations of the mean.



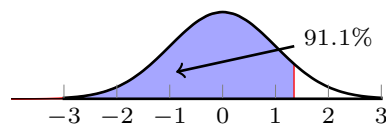
- b. Find the percentage of data lies in each of the intervals.



5. Find the following probabilities,

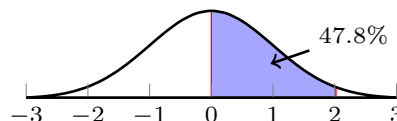
- a. $p(z < 1.35)$

$$p(z < 1.35) \approx 91.15\%$$



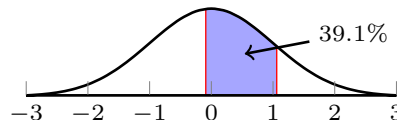
- b. $p(0 < z < 2.02)$

$$p(0 < z < 2.02) \approx 47.83\%$$



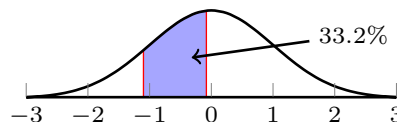
- c. $p(-.09 < z < 1.06)$

$$p(-.09 < z < 1.06) \approx 39.13\%$$



- d. $p(-1.1 < z < -.08)$

$$p(-1.1 < z < -.08) \approx 33.25\%$$

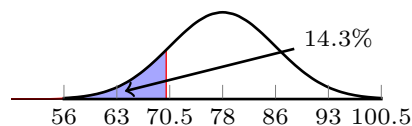


6. A population is normally distributed with a mean 78 and standard deviation 7.5. Find the following probabilities.

a. $p(x < 70)$

$$z_{70} = \frac{70 - 78}{7.5} = -1.0\bar{6} \approx -1.07$$

$$p(z < -1.07) \approx 14.23\%$$

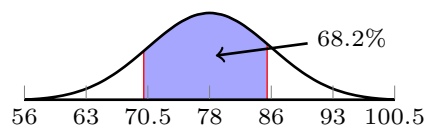


b. $p(70 < x < 85)$

$$z_{70} \approx -1.07 \text{ and } z_{85} = \frac{85 - 78}{7.5} \approx .93$$

$$p(-1.07 < z < .93)$$

$$= p(z < .93) - p(z < -1.07) \approx 68.15\%$$

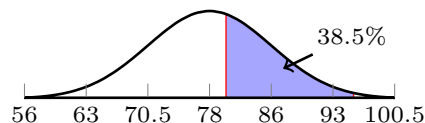


c. $p(80 < x < 95.5)$

$$z_{80} = \frac{80 - 78}{7.5} \approx .27 \text{ and } z_{95.5} = \frac{95.5 - 78}{7.5} \approx 2.33$$

$$p(.27 < z < 2.33)$$

$$= p(z < 2.33) - p(z < .27) \approx 38.37\%$$

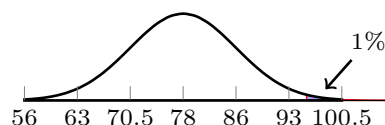


d. $p(95.5 < x < 108)$

$$z_{95.5} \approx 2.33 \text{ and } z_{108} = \frac{108 - 78}{7.5} = 4$$

$$p(2.33 < z < 4)$$

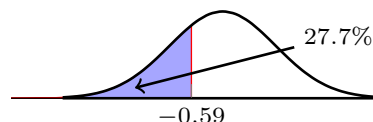
$$= p(z < 4) - p(z < 2.33) \approx 0.99\%$$



7. Find c such that each of the following is true.

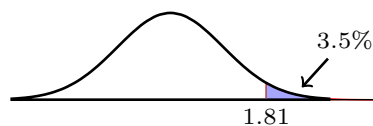
a. $p(z < c) = 0.2776$

$$\begin{array}{l} .2776 \rightarrow \text{area} \\ c = -.59 \end{array} \quad \begin{array}{l} \text{z-score} \\ \end{array}$$



b. $p(z > c) = .0351 \leftarrow \text{area in right tail}$

$$\begin{array}{l} \text{Find z-score for } 1 - .0351 = .9649 \\ .9649 \rightarrow \text{area} \\ c = 1.81 \end{array} \quad \begin{array}{l} \text{z-score} \\ \end{array}$$

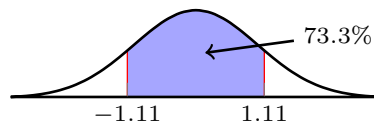


c. $p(-c < z < c) = .733$

The area is split on either side of the mean,

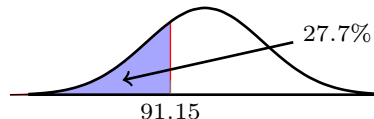
$$\text{Area} = \frac{.733}{2} = .3665$$

$$\begin{array}{l} \text{area} \\ c = 1.11 \end{array} \rightarrow \begin{array}{l} \pm 1.11 \\ \text{z-score} \end{array} \text{ (using Appendix F)}$$



- d. Let $\mu = 100$ and $\sigma = 15$ find c such that $p(x < c) = 27.76\%$

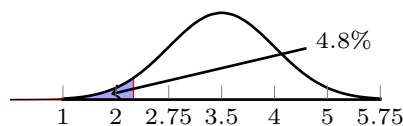
$$\begin{array}{l} \text{area} \\ c = 91.15 \end{array} \rightarrow \begin{array}{l} -.59 \\ \text{z-score} \end{array} \quad \begin{array}{l} c = \sigma z + \mu \Rightarrow c = 15(-.59) + 100 \\ c = 91.15 \end{array}$$



8. The time it takes latex paint to dry is normally distributed with $\mu = 3.5$ hours and $\sigma = .75$ hours. Find the probability that drying time will be as follows.

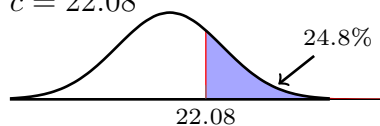
- a. less than 2.25 hours.

$$\begin{array}{l} z_{2.25} = \frac{2.25 - 3.5}{.75} = -1.6\bar{6} \approx -1.67 \\ p(x < 2.25) = p(z < -1.67) = 4.75\% \end{array}$$



9. The average score on the ACT is 18 with a standard deviation of 6. If a college only wants to accept students with an ACT score at least in the top 25%. What is the minimum score they should accept? What about the top 5%?

top 25%: $\begin{array}{l} \text{area} \\ c = 22.08 \end{array} \rightarrow \begin{array}{l} .75 \rightarrow \approx .67 \\ \text{z-score} \end{array}$
 $c = \sigma z + \mu \Rightarrow c = 6(.68) + 18$
 $c = 22.08$



top 5%: $\begin{array}{l} \text{area} \\ c = 27.9 \end{array} \rightarrow \begin{array}{l} .95 \rightarrow \approx 1.645 \\ \text{z-score} \end{array}$
 $c = \sigma z + \mu \Rightarrow c = 6(1.65) + 18$
 $c = 27.9$

